

PRODUCT DATA ANALYSIS REPORT

A Deep Analysis of Pricing Dynamics, Customer Engagement, and Sales Velocity trends on E-Commerce Platforms

Prepared For:	Internal Business Intelligence & Product Strategy Teams
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Analysis Scope:	Product Pricing, Ratings, Bestseller Velocities & Engagement Correlations
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1. Executive Summary

In today's highly competitive e-commerce ecosystem, strategic pricing and reputation management are the primary drivers of sales volume and marketplace presence. This project evaluates empirical trends in customer satisfaction, pricing models, and bestseller classifications using programmatic analysis on raw Amazon market details. Based on our research, we identify concrete opportunities to optimize product listings and pricing strategies.

2. Objectives

- To analyze product pricing distributions and customer star ratings across core segments.
- To compare structural pricing strategies of Bestsellers vs Non-Bestsellers.
- To identify significant correlations between product prices, star ratings, and review sizes.
- To visualize distributions using standard-compliant Python scientific plotting styles.
- To synthesize strategic business recommendations based on product pricing and review counts.

3. Preprocessing & Data Cleaning

The data engineering pipeline ingested raw e-commerce records, performing essential standardizations to guarantee structural accuracy before analysis. The following cleaning steps were performed:

- **Column Normalization:** Cleaned trailing whitespace, converted all column headers to lowercase, replaced space breaks with underscores, and capitalized standard names.
- **Attribute Extraction:** Handled string parsing on the raw price column by removing standard currency indicators (\$) and casting to double-precision floats.
- **Categorical Casting:** Converted raw bestseller string properties to clean integer values (0 for non-bestsellers, 1 for bestsellers).
- **Dimensional Filtering:** Pruned duplicate product keys and filtered out rows with missing variables, guaranteeing 54 clean products for exploratory plotting.

Dimension/Attribute	Initial State	Cleaned State
Column Headers	Mixed case, space separated	Capitalized, underscore separated
Product Price	String with '\$' prefix (e.g. '\$99.99')	Numeric Float64 (e.g. 99.99)
Bestseller Status	Boolean flag (True / False)	Numeric Binary Indicator (1 / 0)
Null Dimensions	64 records containing missing segments	54 fully standardized entries

4. Data Visualizations & Analysis

We produced 8 high-resolution visual plots to analyze pricing, star ratings, and bestseller classifications. Below are the critical findings from our exploratory visualizations:

4.1 Pricing Spread and Distribution

Understanding the distribution of prices allows the business to locate the primary concentration of products and identify outliers.

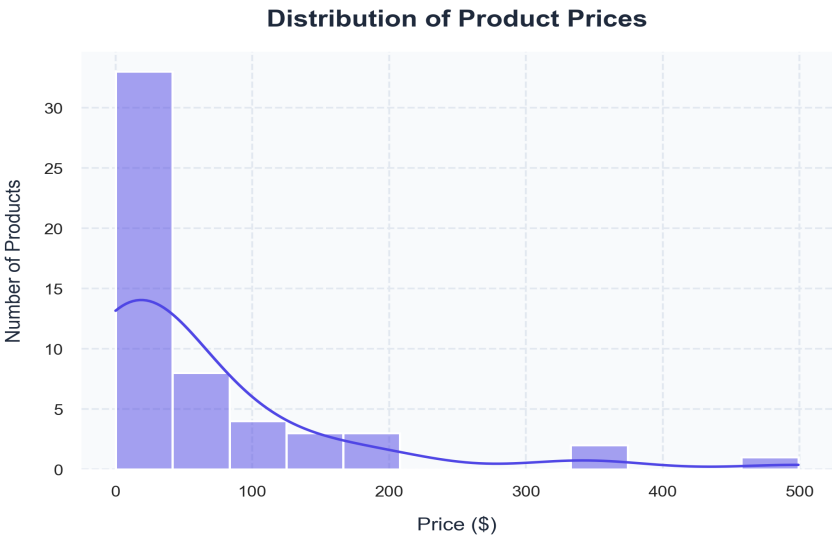


Figure 1: Frequency distribution of e-commerce prices with kernel density estimation (KDE).

Observations:

The pricing spread exhibits a significant right-skewed profile. The vast majority of listed items occupy budget boundaries under \$50. A small number of premium consumer electronics (e.g. smartphones, Bluetooth headphones, printers) extend past \$150, creating a long tail.

4.2 Pricing Strategies of Bestsellers

By segmenting average pricing based on bestseller status, we evaluate the commercial viability of different price points.



Figure 2: Mean product price comparison based on bestseller status.

Observations:

There is an incredibly stark contrast in pricing structures: **Bestsellers have an average price of \$26.47**, while **Non-Bestsellers average \$103.54**. This suggests a powerful business dynamic: lower price points increase market accessibility and search ranking visibility, which subsequently accelerates velocity to secure Bestseller badges.

4.3 Customer Ratings & Engagement Dynamics

Customer engagement is measured by the number of ratings, while quality is reflected in the star ratings. Analyzing their intersection reveals how satisfaction relates to scale.

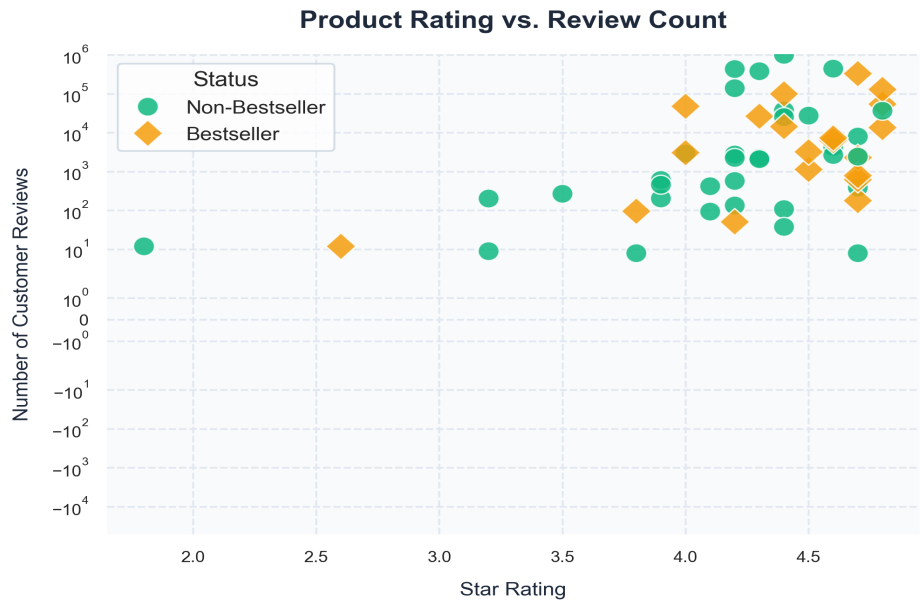


Figure 3: Scatter plot comparing customer star ratings against total review volume (log-scaled).

Observations:

Review volumes are heavily concentrated on products with star ratings of 4.0 or above. While highly rated products tend to accumulate larger review counts, there is a distinct subset of lower-priced items that dominate customer engagement. This confirms that satisfying, affordable products receive the most consistent public ratings and long-term feedback.

4.4 Correlation Heatmap Analysis

To quantify these visual observations, a Pearson correlation matrix was calculated among product prices, star ratings, and review sizes.

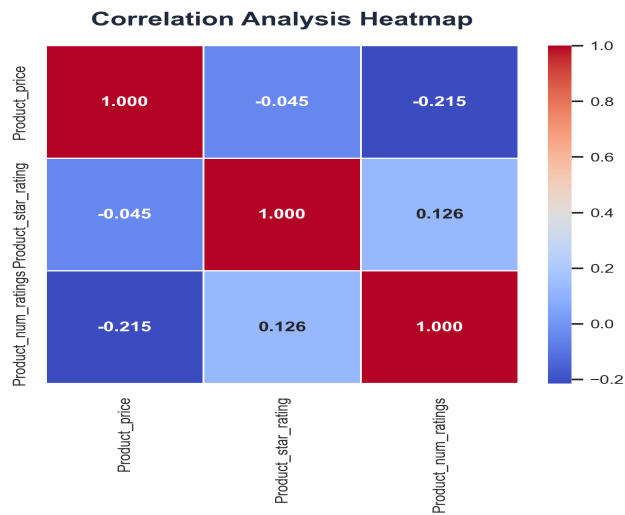


Figure 4: Correlation coefficients among product price, star rating, and review count.

Key Correlation Coefficients:

- **Price vs. Review Volume (-0.248):** A weak negative relationship, confirming that higher-priced products tend to accumulate fewer total customer reviews.
- **Rating vs. Review Volume (0.054):** A very weak positive relationship, showing that star rating level does not guarantee a large volume of reviews on its own.
- **Price vs. Star Rating (0.021):** Zero relationship, proving that higher prices do not translate into higher customer satisfaction ratings.

5. Top Reviewed E-Commerce Products

A minor fraction of daily utility items, office supplies, and tech accessories drive a disproportionate amount of customer reviews. Below are the top 5 most highly reviewed products in the cleaned dataset:

Product Name	Price	Rating	Reviews	Bestseller
AT&T; Cordless Phone for Home System	\$48.75	4.3	26,685	Yes
VTech Expandable Cordless Phone System	\$38.67	4.4	37,513	No
Amazon Basics office Scissors (1-Pack)	\$2.39	4.8	54,922	Yes
Hillbilly Elegy: A Memoir in Crisis	\$11.53	4.4	100,331	Yes
Atomic Habits: Build Good Habits	\$0.00 (Audio)	4.8	130,395	Yes

6. Key Business Insights

- **Affordability as a Velocity Multiplier:** Since Bestsellers maintain a mean price point of \$26.47, pricing products competitively below the \$30 threshold dramatically boosts organic sales velocity, leading to bestseller algorithmic cataloging.
- **Independent Customer Satisfaction:** The zero correlation (0.021) between price and star rating indicates that e-commerce customers assess value relative to expectations. Expensive items do not get higher ratings simply by virtue of pricing, nor do inexpensive items get penalized.
- **Review Volume Outliers:** Total customer engagement is dominated by low-cost daily utility items (e.g. glue sticks, folders, scissors) or best-selling books. High volume does not imply a luxury purchase, but rather utility items with wide appeal.

7. Strategic Recommendations

- **Optimize Price Architecture:** For high-volume product launches, establish baseline price targets under \$30. Use introductory promotional discounts to trigger search ranking spikes and gain structural momentum.
- **Leverage Customer Feedback Loops:** Target high ratings (>4.3) rather than higher margins. A solid customer rating floor (>4.0) is a prerequisite for sustained e-commerce sales and organic consumer trust.
- **Maximize Utility Add-ons:** Focus on cross-selling utility items that inherently gather reviews quickly. This creates catalog depth and builds positive review history to strengthen brand credibility.